

Volatility linkages across three major equity markets

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Defining Market Exuberance

A mathematical measure of market panic or euphoria, totally cleared from economic fundamentals

$$\left(\frac{\Delta s_{t+1}}{s_t} - i_{Lt} \right) = \alpha + \beta \left(\frac{\Delta F_{Lt+1}}{F_{Lt}} \right) + u_{t+1}$$

Left side - excess stock return over bond yield = Right side - changes in bond futures

Arbitrage benchmark: Long-term government bonds (cause assumed to be bubble-free)

The Index $u(t+1)$ - the unexplained residual; represents pure irrational investor behavior (exuberance)

GARCH Methodology

$$h_{jj,t+1} = a_0 + a_1 u_{jt}^2 + a_2 h_{jj,t} + \gamma S_t u_{jt}^2$$

Why GARCH? Financial returns show strong "volatility clustering" (turbulent days follow turbulent days)

Key assumption: "Bad news" generates a significantly larger impact on market volatility than "good news" of the same magnitude

Objectives & Datasets

- USA

- DJI



- UK

- FTSE 100



- Germany

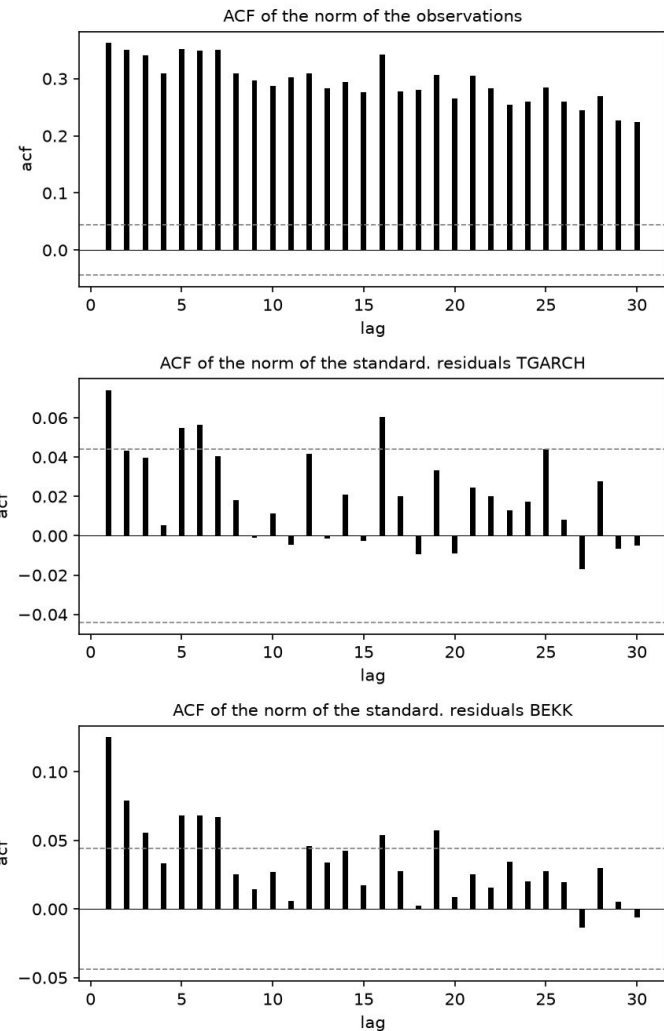
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- Main hypothesis: International crisis contagion is driven by exuberance, not just fundamental stock returns.
- Time period: 01.1992 – 04.2000.
- Variables:
 - Daily stock market returns;
 - 10-year government bond yields;
 - Bond futures settlement prices.

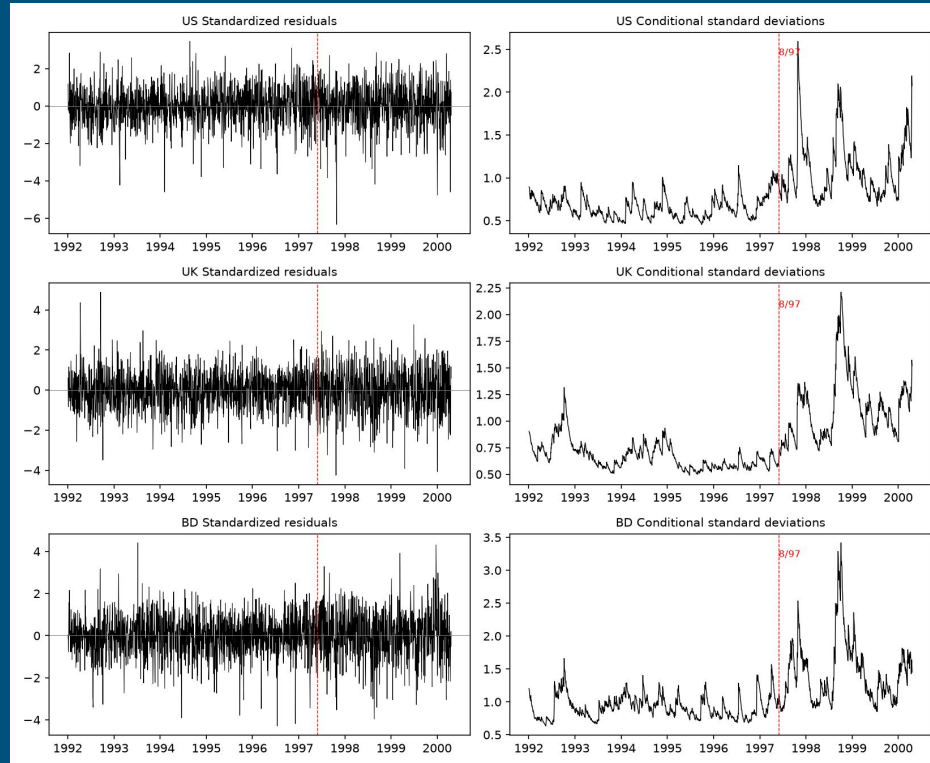
Observation 1: model fit

- Raw Data Issue: The top panel shows massive serial correlation in the raw data (volatility clustering is obvious).
- TGARCH Superiority: The middle panel (TGARCH residuals) shows almost no significant autocorrelation left.
- Conclusion: The chosen TGARCH model perfectly filters the data and captures the non-linear dynamics better than alternative models (like BEKK).



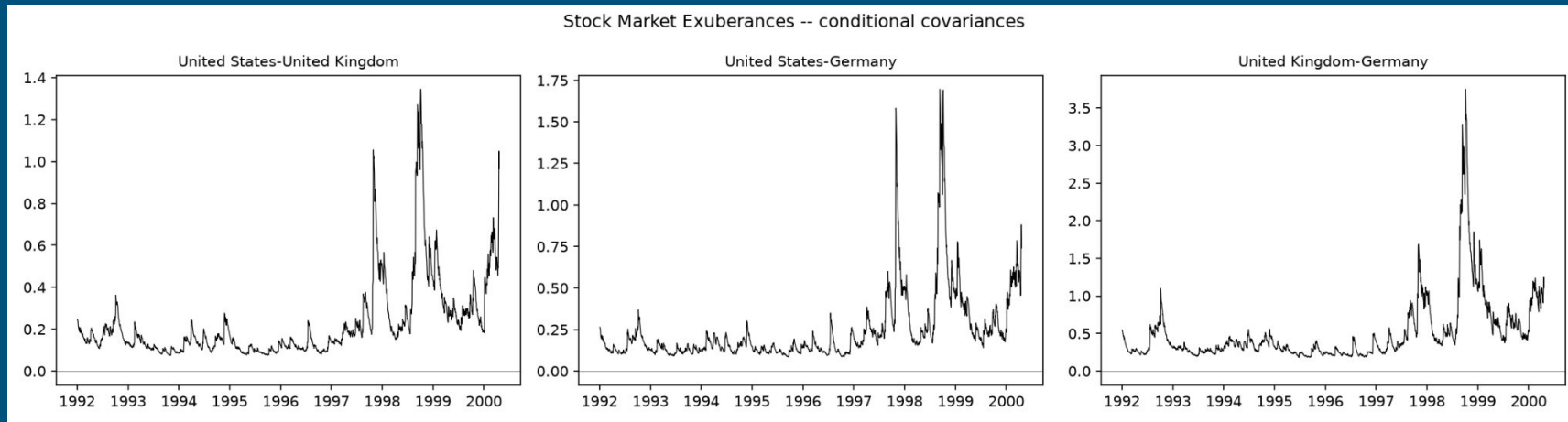
Observation 2: structural shift

- The Regime Shift: A massive, permanent shift in global market volatility occurred in Summer 1997.
- Trigger: The inception of the Asian Financial Crisis.
- Visual Evidence: Conditional standard deviations (right panels) violently spike after the red "8/97" line across all three developed markets.
- Result: Justifies splitting the analysis into "pre-crisis" (calm) and "post-crisis" (turbulent) periods.



Observation 3: contagion

- Measuring Linkages: The charts show how synchronously panic spreads between US, UK, and Germany.
- Crisis Peaks: Massive spikes align perfectly with global shocks:
 - October 1997: Hong Kong market crash.
 - August 1998: Russian debt default.
- The Insight: In turbulent times, cross-market covariances are mostly driven by exuberance (herd behavior), completely ignoring the stable domestic economies.



Conclusion: market exuberance drives volatility contagion

WHAT THE PAPER ARGUES

Volatility transmission is best captured by "market exuberance" rather than raw stock returns.

A structural regime shift in global market volatility occurred in Summer 1997 (Asian Crisis).

Cross-market linkages intensify during crises due to irrational herd behavior (flight to quality).

OUR ASSESSMENT

TGARCH residuals confirm that exuberance indices filter out noise better (removing autocorrelation).

Reconstructed conditional variances show massive, coordinated spikes post-August 1997 across all markets.

Extreme covariance peaks precisely match the 1997 Hong Kong crash and 1998 Russian default.